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Studierichting: Informatica

Oefeningenreeks 4A nr. 18.8

$$\begin{aligned}\int x \sin^2 x dx &= \int x \cdot \frac{1 - \cos 2x}{2} dx = \frac{1}{2} \int x - x \cos 2x dx \\&= \frac{1}{2} \int x dx - \frac{1}{2} \int x \cos 2x dx = \frac{x^2}{4} - \frac{1}{4} \int x \cos 2x d(2x) \\&= \frac{x^2}{4} - \frac{1}{4} \int x d(\sin 2x) = \frac{x^2}{4} - \frac{1}{4} x \sin 2x + \frac{1}{4} \int \sin 2x dx \\&= \frac{x^2}{4} - \frac{1}{4} x \sin 2x + \frac{1}{8} \int \sin 2x d(2x) \\&= \frac{x^2}{4} - \frac{1}{4} x \sin 2x - \frac{1}{8} \cos 2x + c\end{aligned}$$

## References